

BARSAT KHADKA

CONTACT

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EDUCATION

The University of Southern Mississippi
Hattiesburg, MS
B.S., Computer Engineering GPA: 3.92

2024 – PRESENT

FELLOWSHIPS & PROGRAMS

Break Through Tech AI/ML

2026 – 2027

Selected participant, [Cornell Tech] — applied machine learning fellowship.

San Diego Supercomputer Center CIML

JUNE 2026

Summer Institute, UC San Diego — selected for funded HPC/ML training institute.

UR2PhD — Computing Research Association

JULY 2026

Undergrad Research to PhD Program.

PUBLICATIONS

CTS-Bench: Benchmarking Graph Coarsening Trade-offs for GNNs in Clock Tree Synthesis

B. Khadka, K. Roxy, M.R. Ahmed

ACCEPTED, MLBench Workshop @ ASPLOS 2026

arxiv.org/pdf/2602.19330

PREPRINTS & UNDER REVIEW

ChipDreamer: Towards a World Model for Chip Physical Design

B. Khadka, K. Roxy, M.R. Ahmed

ONGOING

SwiftCTS: Fast Cross-Design Prediction and Pareto Optimization of Clock Tree Metrics via Few-Shot Calibration

B. Khadka, K. Roxy, M.R. Ahmed

UNDER REVIEW, ACM TDOAES

arxiv.org/pdf/2606.11348

Filter then Verify: A Multiphase Social Engineering Attack Detection Framework using GNN and BERT for email networks

B. Khadka, P. Koirala, K. Neupane, N. Rahimi

UNDER REVIEW, Springer

arxiv.org/pdf/2605.17201

MechRL: Reinforcement Learning Agents Perform Circuit Discovery for Mechanistic Interpretability

B. Khadka (single author)

UNDER REVIEW, IEEE TAI

arxiv.org/pdf/2605.26343

RESEARCH EXPERIENCE

Machine Learning for Chip Design

2025 – PRESENT

Student Researcher

ADVISOR: Dr. Rubel Ahmed, Louisiana Tech (Remote)

- Built CTS-Bench, a benchmark suite evaluating graph-coarsening trade-offs in graph neural networks applied to VLSI clock tree synthesis.
- Designed SwiftCTS, a physics-informed gradient-boosted surrogate (XGBoost/LightGBM)

that trains in under 5 s on CPU with sub-millisecond inference; a K-shot calibration scheme cuts power error from 24.5% to 3.3% and wavelength error from 56.6% to under 1% on out-of-distribution macros.

- Led ideation, experimental design, ablation studies, manuscript preparation and ran experiments

Machine Learning for Cybersecurity

AUG. 2025 – PRESENT

Cyber Innovation Lab, USM

ADVISOR: Dr. Nick Rahimi

- Proposed a two-stage “filter-then-verify” framework using inductive Graph Neural Networks (GraphSAGE) and ModernBERT to detect multi-stage social engineering attacks.
- Built a three-phase structural anomaly pipeline (volume, relational, and contextual) to flag anomalous sender–receiver patterns in dynamic email networks.
- Reached 92% precision by adding a co-attention ModernBERT verifier for semantic intent, cutting false positives on the Enron dataset.

Research Software Engineer

NOV. 2025 – PRESENT

EDT (NOAA Research Agent)

Institute for Advanced Analytics & Society, USM

- Built EDT, an “analyst-at-scale” research agent that quantifies and communicates the downstream impact of NOAA oceanographic and environmental data products on public-sector decision-making.
- Engineered a multi-provider model backend (Claude Sonnet 4.6 / GPT-4.5) with Model Context Protocol (MCP) tools querying federal and academic databases, all running inside single-tenant Daytona sandboxes.
- Implemented deterministic skill packs (citation resolution, reference merging, REACH impact scoring, multi-agent research orchestration) so quantitative steps run as real Node.js/Python rather than LLM math.

INDUSTRY EXPERIENCE

Software Engineering Intern (Backend)

JUN. 2025 – AUG. 2025

Sports Media Inc (Dakdan Worldwide) Dover, DE

- Built Spring Boot microservices and REST APIs to support real-time ad-bidding workflows.
- Integrated Apache Kafka to power live inventory updates across 100+ venues.
- Contributed to CI/CD pipelines using Jenkins, Maven, and Git; deployed services on AWS and GCP, and wrote JUnit test suites in a Unix/Linux environment.

HONORS & SCHOLARSHIPS

Academic Excellence Award — Full Tuition (4 years)

AUG. 2024

Freshman Designation Valedictorian Scholarship

AUG. 2024

Honors Scholar Recognition

AUG. 2024

1st Runner-Up, VOXO Hackathon

NOV. 2024

Reclaiming Time with Generative AI

President’s List

SPRING 2025

SPECIAL PROJECT	Vinaya Journal <i>Spring Boot, FastAPI, React, SQLite, Chroma, Electron.js</i>	2024 – PRESENT Live · GitHub
	• Built a privacy-first, full-stack desktop journaling app integrating a local LLM + RAG pipeline for offline AI journaling. • Implemented RESTful Spring Boot endpoints with embedded SQLite, plus a FastAPI microservice with Chroma for semantic/vector search, LLM interactions via Ollama, and mood classification. • 35+ active users; 150+ GitHub stars; available for Windows and Linux.	
TALKS & TUTORIALS	Positive Uses, Negative Uses, and Limitations of Virtual Reality <i>MAS 89th Annual Conference, Biloxi, MS</i>	2024
	• Presented findings on VR technology and VRET applications. • Collaborated with peers Anish Kharal and Nhoojah Maharjan in CSC 309; supported by Dr. Sengupta.	
TECHNICAL SKILLS	PROGRAMMING: Python, Java, JavaScript/TypeScript, SQL, C++ ML / RESEARCH: PyTorch, HuggingFace Transformers, PEFT, WeightWatcher, LM-Eval-Harness, NumPy, pandas, Chroma FRAMEWORKS: React, Spring Boot, FastAPI, Maven, Tailwind CSS, PostgreSQL TOOLS & DEVOPS: Git, GitHub, Docker, AWS, GCP, Apache Kafka, Jenkins, JUnit, Postman LANGUAGES: English (fluent), Nepali (native), Hindi (advanced)	
RESEARCH INTERESTS	Reinforcement Learning, Mechanistic interpretability, Physical Design, World Models, AI for Physical Design, ML in Cybersecurity	